Lesson 2: Key Press Reactor

TIC-80

VIDEO GAME CREATION

This week: sequences, events, and the game loop. Your job is to build something that reacts when you press a key - and this is also the start of your Semester 1 project, so save it somewhere safe and give it a name you'll recognize.

Tip: this lesson's Assets folder already has a ready-to-use placeholder character image (and an optional sound effect for Level 3) - you don't need to find or make your own art today.

Setup

1. Launch TIC-80.
2. At the console, type new lua and press Enter to start a fresh cart.
3. Open the Sprite Editor (paintbrush icon) and draw something simple on sprite index 1 - this will be your character (feel free to use one of the images in this lesson's Assets folder as a reference to copy, since TIC-80 draws sprites in its own tiny built-in editor rather than importing files directly).

Level 1 - Minimal (MVP)

One key press causes one visible reaction. This is a complete finish for today - reaching this is enough to move on to next week.

4. Press Esc to reach the code editor.
5. Type the code below, replacing any default code.
6. Press Esc, type run, press Enter, then hold the Right arrow key to test.

```
x=60 y=60
function TIC()
 cls(0)
 if btn(3) then x=x+1 end
 spr(1,x,y,0)
end
```

Level 2 - Complete (Alpha/Beta)

All 4 arrow keys move your character in 4 directions.

- Add the other 3 directions using btn(0) for up, btn(1) for down, and btn(2) for left (adjust x or y accordingly).

Level 3 - Refined (Ready to Publish) - pick any that interest you

- Animate: make your sprite "blink" while moving and stay static when idle. Alternate the scale argument of spr() between 0 and 1 using a frame counter (a scale of 0 draws nothing at all) - this also means switching to spr()'s full argument list, including w/h for a bigger, 2x2-tile composite sprite if you want one.
- Sound: add a beep-beep while moving, silent when idle. Open the SFX Editor via the speaker icon (or press F4) - the musical note icon opens the separate Music Editor instead, which is not what you want here. Alternate between two short SFX slots for a beep-beep while a direction key is held, and call sfx(-1) the instant the key is released - a plain sfx() call has no built-in "stop," so you have to turn it off yourself.
- Recolor: use a different color for each direction, returning to the original color when idle. Temporarily swap one of your sprite's palette colors using poke4() - open the Sprite Editor (F2, or the monster-icon button) to see the color palette strip and figure out which color index is worth changing; there's no pal()/palette() function in TIC-80 like some other fantasy consoles have. Unlike every other engine in this lesson (which swap in a whole separate colored image for this goal, since their tint functions only look clean on white art), TIC-80's poke4() genuinely recolors in place with no such catch - simpler here, not a workaround.
- Clamp movement so the sprite can't leave the 240x136 screen.
- Structure: once you're combining more than one of these, fold the new logic into the same if/elseif/else block from Level 2 rather than writing a separate block for each - see the Key for a worked example of how movement, blink, recolor, and sound can all live together in one place.

If Something's Wrong

- Nothing moves: double-check the function is named exactly TIC() (capitalization matters), and that you pressed run after editing.
- Sprite looks blank or wrong: make sure you drew something on sprite index 1 in the Sprite Editor first.

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tic80.com, Student Resource List > TIC-80